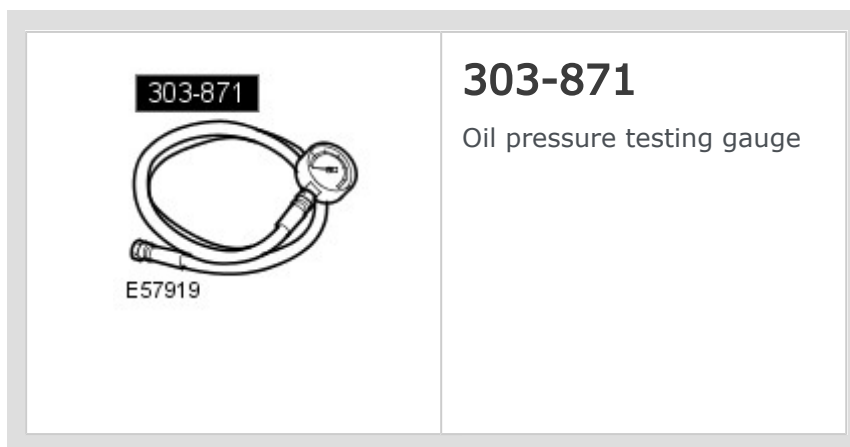


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2015.0 RANGE ROVER (LG), 303-00

ENGINE SYSTEM - GENERAL INFORMATION

SPECIAL TOOL(S)



PRINCIPLE OF OPERATION

For a detailed description of the 4.4L Diesel engine, refer to the relevant Description and Operation section in the workshop manual.

REFER to: [Engine](#) (303-01E Engine - TDV8 4.4L Diesel, Diagnosis and Testing).

INSPECTION AND VERIFICATION

1. Verify the customer concern.

1. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Coolant level ▪ Coolant leaks ▪ Oil level ▪ Oil leaks ▪ Visibly damaged or worn parts ▪ Loose or missing nuts or bolts	▪ Wiring harness ▪ Electrical connector(s) ▪ Injectors ▪ Glow plugs ▪ 5 volt sensor supply ▪ Sensor(s) ▪ Cooling fan control module and motor ▪ Engine Control Module (ECM)

2. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

1. If the concern is not visually evident, verify the symptom and refer to the relevant Symptom Chart. Symptom Charts have been separated into **Leaks** and **Noise Vibration and Harshness (NVH)** for ease of use. Alternatively, check for Diagnostic Trouble Codes (DTCs) and refer to the relevant DTC Index.

SYMPTOM CHARTS

Symptom Chart, Leaks

SYMPTOM	POSSIBLE CAUSE	ACTION
External coolant leaks	▪ Damaged hose(s) ▪ Damaged expansion tank ▪ Damaged radiator ▪ Leaking seals /gaskets ▪ Cracked /damaged casings	For cooling system checks, REFER to: Engine Cooling (303-03F Engine Cooling - TDV8 4.4L Diesel, Diagnosis and Testing).

Internal coolant leaks Note: This may be indicated by the production of white smoke from the exhaust	▪ Leaking seals /gaskets ▪ Cracked /damaged casings	For cooling system checks, REFER to: Engine Cooling (303-03F Engine Cooling - TDV8 4.4L Diesel, Diagnosis and Testing).
Engine overheats	▪ Insufficient coolant ▪ Insufficient oil ▪ Pressure cap fault ▪ Thermostat not opening ▪ Coolant pump failure ▪ Cooling fan failure	For cooling system checks, REFER to: Engine Cooling (303-03F Engine Cooling - TDV8 4.4L Diesel, Diagnosis and Testing).
Engine takes too long to reach operating temperature	▪ Thermostat stuck open	For cooling system checks, REFER to: Engine Cooling (303-03F Engine Cooling - TDV8 4.4L Diesel, Diagnosis and Testing).
External oil leaks	▪ Gaskets ▪ Seals ▪ Oil pipes ▪ Oil filter ▪ Oil cooler ▪ Damaged /cracked casings ▪ Crankcase ventilation system ▪ Piston ring blow-by	Clean and confirm the area of the leak. Check the visual condition of oil carrying components. Check the crankcase ventilation system. Carry out a compression test, GO to Pinpoint Test A .
Internal oil leaks (leaks into coolant or combustion chamber) Note: This may be indicated by the production of blue smoke from the exhaust	▪ Gaskets ▪ Seals ▪ Damaged /cracked casings ▪ Worn valve guides ▪ Worn cylinder bores /pistons	Check for traces of oil in the coolant. Check for evidence of oil in the combustion chambers (deposits on the glow plugs, etc). Confirm oil consumption and vehicle usage with the owner /driver. Carry out an oil consumption test, GO to Pinpoint Test B .

- Broken piston rings

Symptom Chart, NVH



NOTE:

As the checks suggested here are open to interpretation, they should be used as a guide only. Descriptions of noises, etc, are in general terms, so depend on a degree of experience on the part of the technician.

SYMPTOM	POSSIBLE CAUSE	ACTION
Rattle/ticking from top of engine	▪ Valve gear noise ▪ Camshaft bearing noise ▪ Camshaft chain noise ▪ Tensioner noise ▪ Vacuum pump noise ▪ High pressure fuel pump noise	Check the engine oil pressure. Check the function of the hydraulic tappets and the camshaft condition. Check the camshaft bearings, chains and tensioners. Check the vacuum pump, and high pressure fuel pump
Growl from top of engine	▪ High pressure fuel pump belt noise ▪ High pressure fuel pump belt tensioner noise	Check the high pressure fuel pump belt and tensioner
Squeaking /Creaking/Squeal from front of engine	▪ Front End Accessory Drive (FEAD) belt ▪ FEAD belt tensioner ▪ Driven components on FEAD belt	Check the FEAD belt and driven components, REFER to: Accessory Drive (303-05D Accessory Drive - TDV8 4.4L Diesel, Diagnosis and Testing).
Whine/Slap		Check the FEAD belt and driven components. Check the

/Growl from front of engine	▪ Front End Accessory Drive (FEAD) belt ▪ FEAD belt tensioner ▪ Driven components on FEAD belt ▪ Timing belt noise ▪ Timing belt tensioner noise	timing belt and tensioners
Knock from lower half of engine (often worse with a cold engine)	▪ Piston slap ▪ Piston pin noise ▪ Connecting rod bearing noise	Check the engine oil pressure. Check piston, cylinder bore, piston pin and connecting rod bearing for excess wear
Knock/Rumble from lower half of engine (often worse on overrun)	▪ Main bearing noise	Check the engine oil pressure. Check piston, cylinder bore, piston pin and connecting rod bearing for excess wear
Misfire/Rough running	▪ Engine management system ▪ Fuel charging and controls ▪ Exhaust gas recirculation (EGR) system ▪ Burnt /sticking valves ▪ Worn valve guides ▪ Worn cylinder bores/pistons ▪ Broken piston rings ▪ Damaged /cracked casings	For engine management system tests, REFER to: Diagnostic Trouble Code (DTC) Index - TDV8 4.4 L Diesel, DTC: Engine Control Module (ECM) (100-00 General Information, Description and Operation). For fuel charging and controls systems tests, REFER to: Fuel Charging and Controls (303-04F Fuel Charging and Controls - TDV8 4.4L Diesel, Diagnosis and Testing), Turbocharger (303-04G Fuel Charging and Controls - Turbocharger - TDV8 4.4L Diesel, Diagnosis and Testing). For EGR system tests, REFER to: Engine Emission Control (303-08E Engine Emission Control - TDV8 4.4L Diesel, Diagnosis and Testing). Carry out a compression test, GO to Pinpoint Test A. Check for excess wear in engine components

DTC INDEX

For a complete list of all Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.

REFER to: Diagnostic Trouble Code (DTC) Index - TDV8 4.4L Diesel, DTC: Engine Control Module (ECM) (100-00 General Information, Description and Operation).

PINPOINT TESTS



NOTE:

Where reference is made to 'suitable equipment', this refers to standard workshop equipment. Refer to the operating instructions for your own equipment when performing any tests.



WARNING:

PINPOINT TEST A : CHECK THE CYLINDER COMPRESSIONS



WARNING:

Only compression testers able to read the higher compression pressures found in diesel engines should be used. Failure to follow this instruction may result in personal injury.



NOTE:

Where possible, compression testing should be carried out on an engine at operating temperature.

TEST CONDITIONS

DETAILS/RESULTS/ACTIONS

Only compression testers able to read the higher compression pressures found in diesel engines should be used. Failure to follow this instruction may result in personal injury.



NOTE:

Where possible, compression testing should be carried out on an engine at operating temperature.

A1: CARRY OUT A DRY CYLINDER COMPRESSION TEST

- | | | | | | | | |
|---|--|---|---|---|--|---|--|
| 1 Make
sure
the
parking
brake
is
applied
and
that
the
selector
lever is
in park. | 2 Set the
ignition
status
to OFF
. | 3 Remove
the
starter
relay. | 4 Disconnect
the starter
motor
solenoid
connector. | 5 Connect
a
suitable
remote
starter
device
to the
starter
motor
solenoid. | 6 Remove
the
glow
plugs. | 7 Install
adaptor
in place
of the
glow
plug in
the first
cylinder
to be
tested. | 8 Connect a
suitable
compression
tester to the
adaptor.
See warnin
above. |
|---|--|---|---|---|--|---|--|

Are the compression figures within 10% of each other?

Yes

Unless the compression figures are universally very low (experience will indicate this), check for other causes for the customer complaint.

No

[GO to A2 .](#)

A2: CARRY OUT A WET CYLINDER COMPRESSION TEST



CAUTION:

If engine oil is introduced into the cylinders, run the engine at 2,000 rpm for a minimum of ten minutes after completing testing to prevent damage to the catalytic converters. Failure to follow this instruction may result in damage to the vehicle.



NOTE:

There is a combustion chamber in the top of each piston. Make sure that the oil is not allowed to run into this chamber.

- | | | |
|---|---|--|
| 1 Using a suitable oil can with a flexible spout, introduce a small amount of clean engine oil into the cylinder just before testing, such that the oil is able to run between the piston and the cylinder bore. | 2 Repeat steps 7 - 10 from the test above, introducing oil into each cylinder just before testing. | 3 Compare the compression figures across all the cylinders. |
|---|---|--|

Is the compression figure higher than the dry test?

Yes

A higher figure following the introduction of oil may indicate a worn or damaged cylinder bore, piston and/or piston rings. Disassembly would be required to confirm this.

No

If the compression figure is unaffected by the introduction of oil, but the figure is still less than 90% of the other cylinders, this may indicate a burnt and/or sticking valve, leaking head gasket, etc. Disassembly would be required to confirm this.

Clear any DTCs which may have been induced by the test.

 NOTE:	
PINPOINT TEST B : OIL CONSUMPTION TEST	
 NOTE:	
Oil consumption will vary, depending on a number of factors. New engines will normally use more oil than 'run-in' engines, although a guideline would be to expect 16,000 Km (10,000 miles) per liter.	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS

Oil consumption will vary, depending on a number of factors. New engines will normally use more oil than 'run-in' engines, although a guideline would be to expect 16,000 Km (10,000 miles) per liter.

B1: CHECK FOR EXCESSIVE OIL CONSUMPTION

1	Start the engine and allow it to run until it reaches normal operating temperature.	2	Make sure the vehicle is parked on a level surface and set ignition status to OFF .	3	Allow to settle for at least five minutes.	4	Check the oil level.	5	Correct the level, if necessary, and record the reading and mileage in the vehicle history.	6	Make sure that the owner /driver is aware that a test is being carried out, and that they should not top-up their	7	When the oil level reaches the ADD mark, the customer should bring the vehicle in to be checked.	8	Top-up the oil to the level at the beginning of the test and record the amount of oil needed to do so, and the mileage covered in the
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oil level
for the
duration
of the
test,
but
should
check
the
level
every
160-
240 Km
(100-
150
miles).

course of
the test.

Is the consumption excessive for the mileage and/or use?

Yes

Disassembly will be required to check the components indicated in the symptom chart.

No

No further action is required.

ENGINE OIL PRESSURE CHECK



NOTE:

Prior to checking the engine oil pressure, a road test of 6 miles (10 kilometres), must be carried out. Do not attempt to attain engine normal operating temperature by allowing the engine to idle.

1. Disconnect the battery ground cable.

REFER to: [Specifications](#) (414-00 Battery and Charging System - General Information, Specifications).



WARNINGS:

- The spilling of hot engine oil is unavoidable during this procedure, care must be taken to prevent scalding.
- Wear protective gloves.

1. Remove the engine oil pressure sensor.

REFER to: [Engine Oil Pressure Sensor](#) (303-14E Electronic Engine Controls - TDV8 4.4L Diesel, Removal and Installation).

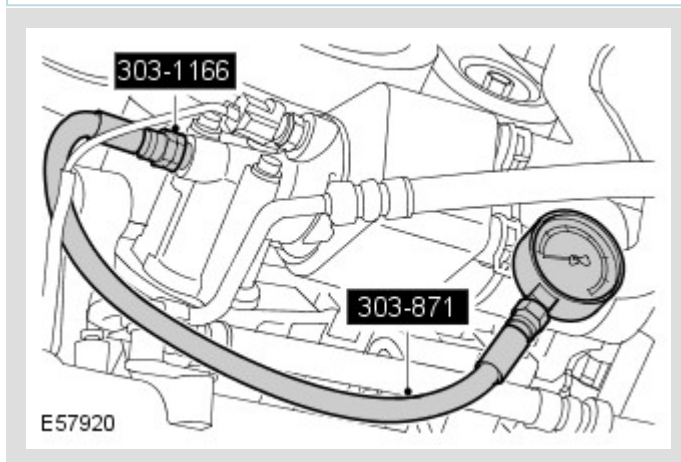
1. Install the special tool.

1. Install the special tool gauge and tighten the union.



NOTE:

4.4L shown.



1. Check and top-up the engine oil if required.

1. Connect the battery ground cable.

REFER to: [Specifications](#) (414-00 Battery and Charging System - General Information, Specifications).

1. Start and run the engine.

1. Note the oil pressure readings with the engine running at idle and 3500 RPM.

REFER to: [Specifications](#) (303-01E Engine - TDV8 4.4L Diesel, Specifications).

1. Turn off the engine.

1. Disconnect the battery ground cable.

1. Remove the special tools.

1. Clean the components.

1. Install the oil pressure sensor.

REFER to: [Engine Oil Pressure Sensor](#) (303-14E Electronic Engine Controls - TDV8 4.4L Diesel, Removal and Installation).

1. Check and top-up the engine oil if required.

1. Connect the battery ground cable.